Sarisht Wadhwa

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Summary

My research develops secure and incentive-compatible mechanisms for decentralized systems, with a particular focus on censorship resistance, transaction ordering, and transaction fee mechanism design on blockchains. I study how cryptographic tools and algorithmic game theory can be combined to address real-world challenges in decentralized finance and distributed protocol design. I have had the privilege to work with wonderful collaborators from Ethereum Research, Flashbots, Category Labs and Common Prefix.

Education

- **Ph.D. in Computer Science.** Duke University. Advisors: *Kartik Nayak* and *Fan Zhang* 2021 – 2026
  Thesis: *Security of Blockchains with Strategic Entities*
- **Dual B.Tech./M.Tech. in Computer Science and Engineering.** IIT Delhi 2014 – 2019

Positions

- **Postdoctoral Researcher.** University of Michigan. Host: *Ke Wu* 2026 –
- **Research Intern.** Flashbots. With *Jonathan Passerat-Palmbach* 2023
- **Senior Software Developer.** Samsung Research Institute, Bangalore. Camera systems 2019 – 2021

Papers

*: equal contribution

9. **Perils of Parallelism: Transaction Fee Mechanisms under Execution Uncertainty** **USENIX Sec'26**
   Authors: Sarisht Wadhwa*, Aviv Yaish*, Fan Zhang, Kartik Nayak
   In: *Proceedings of the 35th USENIX Security Symposium (USENIX Security 2026)*. Also accepted at SBC 2026
8. **Censorship Resistance vs Throughput in Multi-Proposer BFT Protocols** **CCS'26**
   Authors: Fatima Elsheimy*, Ioannis Kaklamanis*, Sarisht Wadhwa*, Charalampos Papamanthou, Fan Zhang
   In: *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS 2026)*
7. **Price of Censorship: Censorship Resistance and Throughput Under Rational Concurrent Proposers** **AFT'26**
   Authors: Aditya Saraf, Ioannis Kaklamanis, Sarisht Wadhwa, Fatima Elsheimy
   In: *8th Conference on Advances in Financial Technologies (AFT 2026)*, LIPIcs
6. **Censorship-Resistant Sealed-Bid Auctions on Blockchains** **2025**
   Authors: Orestis Alpos, Lioba Heimbach, Kartik Nayak, Sarisht Wadhwa (alphabetical order)
   In: *Cryptology ePrint Archive, 2025/2127*
5. **Differentially Private Aggregate Hints in MEV-Share** **2025**
   Authors: Jonathan Passerat-Palmbach, Sarisht Wadhwa
   In: *arXiv:2508.14284*
4. **AUCIL: An Inclusion List Design for Rational Parties** **2025**
   Authors: Sarisht Wadhwa, Julian Ma, Thomas Thiery, Barnabé Monnot, Luca Zanolini, Fan Zhang, Kartik Nayak
   In: *Cryptology ePrint Archive, 2025/194*
3. **Data Independent Order Policy Enforcement: Limitations and Solutions** **CCS'24**
   Authors: Sarisht Wadhwa, Luca Zanolini, Francesco D'Amato, Aditya Asgaonkar, Chengrui Fang, Fan Zhang, Kartik Nayak
   In: *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS 2024)*
2. **He-HTLC: Revisiting Incentives in HTLC** **NDSS'23**
   Authors: Sarisht Wadhwa*, Jannis Stöter*, Fan Zhang, Kartik Nayak
   In: *Network and Distributed System Security Symposium (NDSS 2023)*
1. **Efficiently Answering Regular Simple Path Queries on Large Labeled Networks** **SIGMOD'19**
   Authors: Sarisht Wadhwa, Anagh Prasad, Sayan Ranu, Amitabha Bagchi, Srikanta Bedathur
   In: *Proceedings of the ACM International Conference on Management of Data (SIGMOD 2019)*

Grants and Awards

- **First prize, IC3 Summer Camp** (among eight projects), for an elastic restaking design 2025
- **Ethereum Foundation Academic Grant**, *Censorship Resistance for Ethereum*, with Kartik Nayak 2024
- **Samsung Research Excellence Award**, for research in camera systems 2021
- **Microsoft Research Travel Scholarship**, to attend ACM SIGMOD 2019 in Amsterdam 2019

Service

- **Program Committee**
  - MCP Workshop 2027 (co-located with FC 2027)
  - CAAW 2027 (co-located with FC 2027)
  - USENIX Security 2027
  - ACM CCS 2026
  - ACM EC 2026
  - ACM DeFi Workshop 2026 (co-located with ACM CCS 2026)
  - CAAW 2026 (co-located with FC 2026)
  - Agentic Markets Workshop at ICML 2024
  - ACM DeFi Workshop 2024 (co-located with ACM CCS 2024)
- **External Reviewer**
  - IEEE S&P 2024, 2025, 2026
  - FC 2025, 2026
  - AFT 2022, 2026

Teaching

- **COMPSCI 354: Foundations of Blockchains (TA).** Duke University. Instructor: *Kartik Nayak* Fall 2024
- **COMPSCI 351: Computer Security (TA).** Duke University. Instructor: *Kartik Nayak* Spring 2024
- **COMPSCI 590.1: Foundations of Blockchains (TA).** Duke University. Instructor: *Kartik Nayak* Fall 2022
- **COMPSCI 351: Computer Security (TA).** Duke University. Instructors: *Kartik Nayak* and *Michael Reiter* Spring 2022

Talks

- **AUCIL: An Inclusion List Design for Rational Parties**
  - University of Michigan Theory Seminar Nov 2026
  - Yale Applied Cryptography Lab, invited talk: *Security of Blockchains with Strategic Players* Oct 2025
  - MEV Workshop at the Science of Blockchain Conference (SBC) Aug 2025
- **Perils of Parallelism**
  - IC3 Summer Camp Jun 2026
  - Designing DeFi (D^2) Workshop May 2026
  - Yale Applied Cryptography Lab Apr 2026
  - Columbia Cryptoeconomics Workshop (CCE) Dec 2025
- **Data Independent Order Policy Enforcement**
  - University of Michigan (invited visit) Apr 2026
  - ACM Conference on Computer and Communications Security (CCS) Oct 2024
  - Yale Security Group Jul 2024
  - DeFi Workshop at ACM EC Jul 2024
  - Duke Patent Office Visit Day Jun 2024
- **He-HTLC: Revisiting Incentives in HTLC**
  - Network and Distributed System Security Symposium (NDSS) Feb 2023
  - Crypto Economics Security Conference (CESC) Oct 2022
  - Science of Blockchain Conference (SBC) Aug 2022

Writings

- **AUCIL: An Auction-Based Inclusion List Design for Enhanced Censorship Resistance on Ethereum**, *Ethereum Research*
- **Block Building is Not Just Knapsack**, *Ethereum Research*